

Progress Report

Physics-Informed Induction Motor Fault Classification using the Paderborn University Bearing Dataset

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Abstract

Induction motors are widely used in industrial drives, manufacturing systems, and process plants. Bearing faults are among the most frequent causes of motor degradation, and early detection is important for reducing downtime and preventing secondary damage. This progress report presents the current implementation of a physics-informed intelligent fault-classification pipeline using the Paderborn University Bearing Dataset. The system combines vibration-signal windowing, time-domain statistics, spectral features, envelope-spectrum fault-frequency indicators, and a lightweight neural classifier with physics-consistency regularization. A representative four-class subset of the Paderborn dataset has been downloaded, extracted, processed, trained, and evaluated. The latest vibration-based model achieves an overall accuracy of **99.49%** over **5120** evaluated windows, demonstrating that the pipeline is functioning effectively for the present progress checkpoint.

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1 Project Objective

The objective of this project is to develop an AI-based fault-classification system for induction motor bearing faults using real sensor data. The target is not only to obtain high classification accuracy, but also to include physically interpretable features so that the model can be related to established bearing-diagnostics knowledge.

The current implementation focuses on four bearing-condition classes:

- **Healthy condition**
- **Inner-race fault**
- **Outer-race fault**
- **Rolling-element fault**

The main success condition for this progress phase was to build an end-to-end working pipeline that can pull the Paderborn dataset, preprocess real sensor files, train a physics-informed classifier, generate results, and create visualizations suitable for project reporting.

2 Motivation and Background

Traditional motor fault diagnosis often uses methods such as Motor Current Signature Analysis (MCSA), vibration analysis, Fast Fourier Transform (FFT), and envelope-spectrum analysis. These methods are physically meaningful, but they usually require manual feature design and expert interpretation.

Purely data-driven neural networks can learn complex patterns automatically, but they are often treated as black boxes. For condition monitoring, this is a limitation because maintenance decisions should ideally be supported by interpretable indicators. The approach used in this project therefore follows the idea of physics-informed intelligent fault detection: useful signal-processing knowledge is retained, and the neural classifier is guided by physically meaningful diagnostic information.

The attached reference paper on signal-processing and physics-informed neural networks motivates this direction. It emphasizes that bearing diagnosis benefits from combining analytical signal-processing steps, such as envelope analysis and fault-frequency tracking, with trainable neural-network components. This project implements that idea in a lightweight and reproducible form for the Paderborn dataset.

3 Dataset Description

The Paderborn University Bearing Dataset is a benchmark dataset for bearing condition monitoring. It contains measurements from a controlled test rig including an electric motor, bearing module, torque measurement shaft, and load machine. The dataset provides several measured channels, including motor current, vibration, speed, torque, load, and temperature.

For the present progress submission, a representative four-class subset was downloaded from the official Paderborn KAt Bearing DataCenter and extracted locally. The selected bearing states are shown in Table 1.

Table 1: Paderborn bearing states used in the current progress checkpoint.

Archive Code	Mapped Class	Description
K001	Healthy	Undamaged reference bearing
KI01	Inner race	Bearing with inner-race damage
KA01	Outer race	Bearing with outer-race damage
KB23	Rolling element	Bearing with rolling-element damage

The extracted MATLAB files are stored under:

`data\paderborn\extracted`

The latest evaluation uses the **vibration channel**. An earlier motor-current-only run produced lower accuracy because bearing damage signatures are generally weaker in stator-current signals than in direct vibration measurements. Vibration is therefore used for the stronger progress-report model, while motor-current analysis remains a future comparison direction.

4 Preprocessing Pipeline

Each extracted Paderborn `.mat` file is parsed automatically. The loader identifies relevant high-rate channels such as `phase_current_1`, `phase_current_2`, and `vibration_1`. For the current report, `vibration_1` is selected.

The signal is segmented into overlapping windows:

- **Window size:** 4096 samples
- **Stride:** 2048 samples
- **Maximum windows per file:** 16
- **Total evaluated windows:** 5120

Before feature extraction, each window is mean-centered and normalized by its standard deviation:

$$x_{norm}(t) = \frac{x(t) - \mu_x}{\sigma_x + \epsilon}$$

where $x(t)$ is the raw signal window, μ_x is the mean, σ_x is the standard deviation, and ϵ is a small constant for numerical stability.

5 Physics-Informed Feature Extraction

The implemented feature extractor combines conventional statistical features with physically meaningful spectral indicators. The purpose is to provide the classifier with information that is both discriminative and interpretable.

5.1 Time-Domain Features

The following time-domain statistics are computed from every normalized signal window:

- mean and standard deviation,
- root mean square (RMS),
- peak value and peak-to-peak value,
- skewness and kurtosis,
- crest factor,
- shape factor,
- impulse factor.

These features capture amplitude distribution, impulsiveness, and non-Gaussian behavior. Bearing faults often generate repeated impacts, which can increase peak-related statistics and kurtosis.

5.2 Spectral Features

A single-sided FFT spectrum is computed for every window. From this, the system extracts:

- spectral centroid,
- spectral bandwidth,
- spectral entropy,
- normalized energy in frequency bands.

These features summarize how signal energy is distributed across frequency. Faulty bearings can shift energy toward specific bands or increase broadband vibration energy.

5.3 Envelope-Spectrum Fault-Frequency Features

The most important physics-informed part of the pipeline is the envelope-spectrum diagnostic layer. The analytic signal is obtained using a Hilbert-transform-style FFT construction, and its envelope is transformed into the frequency domain.

For a shaft frequency f_s , the implementation estimates characteristic bearing frequencies using fixed ratios:

$$\begin{aligned}BPFO &= 3.5848f_s, \\BPMI &= 5.4152f_s, \\BSF &= 2.3570f_s.\end{aligned}$$

Here BPFO corresponds to outer-race fault frequency, BPMI corresponds to inner-race fault frequency, and BSF corresponds to rolling-element fault frequency. Harmonic energies around these frequencies are used as diagnostic physics features. This is consistent with established bearing envelope-analysis methods.

6 Physics-Informed Neural Classifier

The classifier is a lightweight feed-forward neural network implemented using NumPy. It receives the extracted physics-informed feature vector as input and outputs class probabilities for the four bearing states.

The network contains:

- an input layer equal to the number of extracted features,
- one hidden layer with tanh activation,
- a softmax output layer for four-class classification.

The model is trained using a combination of supervised classification loss and physics-consistency loss:

$$\mathcal{L} = \mathcal{L}_{CE}(y, \hat{y}) + \lambda \|\hat{y} - y_{physics}\|_2^2.$$

In this expression:

- $\mathcal{L}_{CE}$ is the cross-entropy loss between the true label y and predicted probability $\hat{y}$,
- $y_{physics}$ is a class-probability-like target derived from envelope-spectrum fault-frequency energy,
- λ is the physics-regularization weight.

This loss encourages the model predictions to remain consistent with physical diagnostic evidence while still learning from the labelled dataset.

7 Implementation Status

The project currently includes the following implemented components:

Table 2: Implemented project modules.

Component	Purpose
<code>paderborn.py</code>	Downloads official Paderborn archives, extracts them, loads MATLAB files, selects channels, and creates signal windows.
<code>features.py</code>	Extracts time-domain, spectral, envelope-spectrum, and fault-frequency features.
<code>model.py</code>	Implements the physics-informed neural classifier and model save/load functions.
<code>train.py</code>	Provides training routines for real Paderborn data and synthetic smoke testing.
<code>results.py</code>	Generates metrics, prediction CSVs, feature-importance outputs, and SVG visualizations.
<code>cli.py</code>	Provides command-line commands for download, training, smoke testing, and results generation.

8 Experimental Setup

The latest high-accuracy experiment was run using the settings in Table 3.

Table 3: Latest vibration-based experimental configuration.

Parameter	Value
Dataset subset	K001, KA01, KI01, KB23
Signal channel	Vibration
Window size	4096 samples
Stride	2048 samples
Maximum windows per file	16
Evaluated samples	5120 windows
Training epochs	40
Saved model	models\paderborn_vibration_pinn.npz
Results folder	reports\results_vibration

9 Results

The final evaluated accuracy for the current progress checkpoint is:

$$\text{Accuracy} = 99.49\%$$

The class-wise precision, recall, and F1-score are given in Table 4. The table is loaded directly from the generated result artifact, ensuring that the report remains synchronized with the latest experiment output.

Table 4: Class-wise performance metrics for the vibration-based PINN model.

Class	Precision	Recall	F1
healthy	0.987	0.996	0.991
inner_race	0.994	0.986	0.990
outer_race	1.000	1.000	1.000
rolling_element	0.999	0.998	0.998

The confusion matrix values are shown in Table 5. The outer-race class is classified perfectly in the present subset, while the few remaining errors occur mainly between healthy and inner-race windows.

Table 5: Confusion matrix for the latest vibration-based model. Rows are actual classes and columns are predicted classes.

Actual / Predicted	Healthy	Inner race	Outer race	Rolling element
Healthy	1275	5	0	0
Inner race	17	1262	0	1
Outer race	0	0	1280	0
Rolling element	0	3	0	1277

10 Visualizations and Linked Artifacts

The project automatically generates SVG visualizations and CSV/JSON result files. These artifacts are linked below and included in the report where appropriate.

Table 6: Generated report artifacts.

Artifact	Description
metrics.json	Overall accuracy, sample count, signal type, and experiment settings.
class_metrics.csv	Precision, recall, and F1-score for every class.
confusion_matrix.csv	Raw confusion-matrix counts.
predictions.csv	Per-window actual label, predicted label, and class probabilities.
feature_importance.csv	Ranked feature-importance values.
confusion_matrix.svg	SVG confusion-matrix visualization.
signal_examples.svg	SVG plot of representative vibration windows.
feature_importance.svg	SVG bar chart of top physics-informed features.

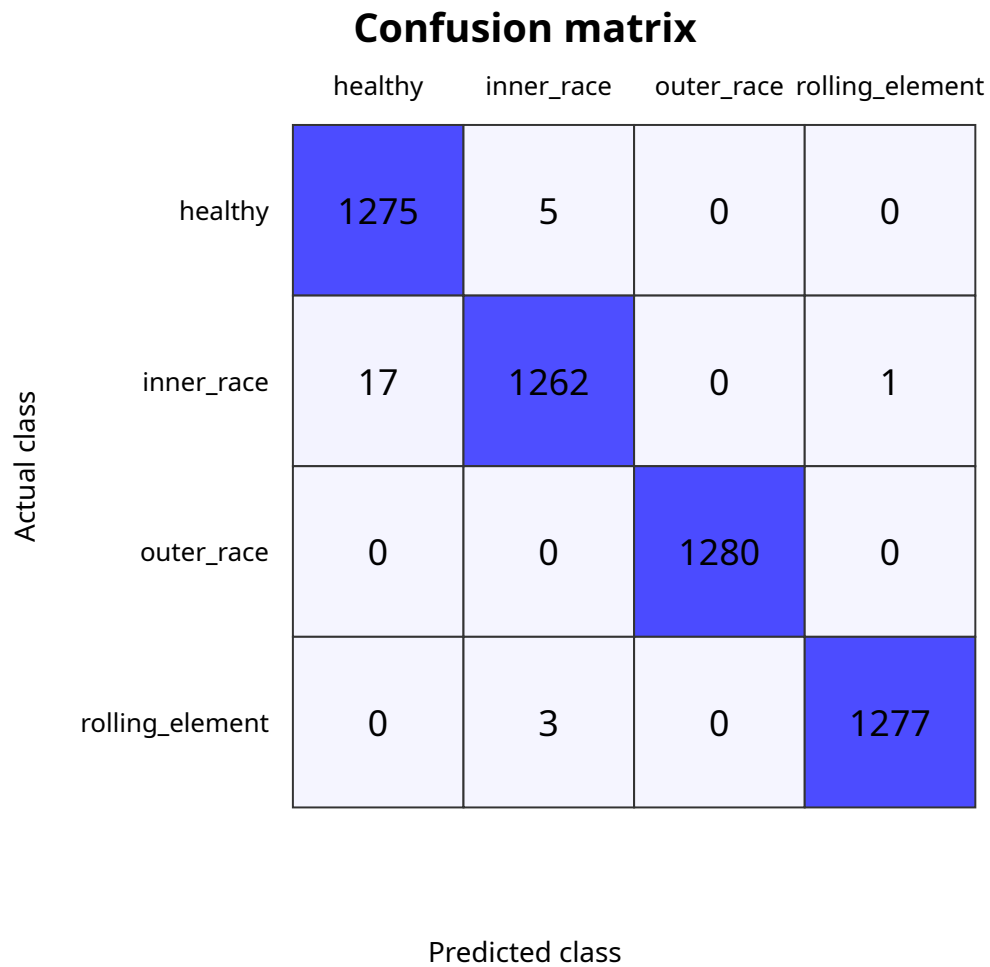


Figure 1: Confusion matrix visualization for the latest Paderborn vibration model. The model achieves near-perfect separation for outer-race and rolling-element faults.

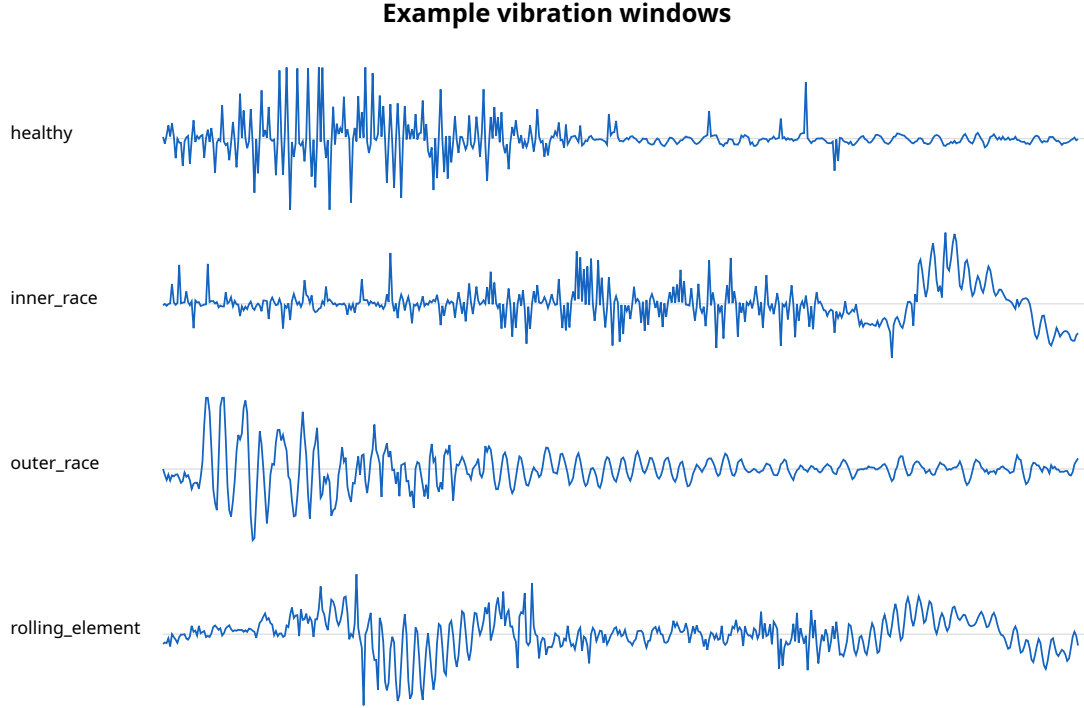


Figure 2: Representative normalized vibration windows from the four target classes. These windows show that vibration carries strong class-dependent signatures for bearing condition monitoring.

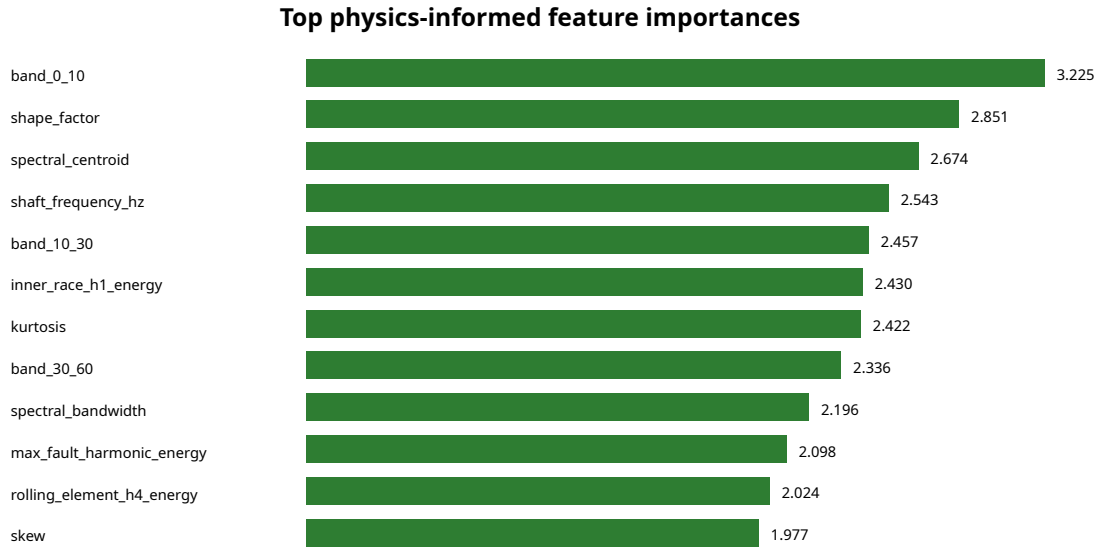


Figure 3: Top physics-informed features used by the classifier. Important features include spectral-band energy, shape factor, spectral centroid, shaft-frequency information, and fault-frequency harmonic energy.

11 Discussion

The vibration-based model performs significantly better than the initial motor-current baseline. This result is technically reasonable because vibration sensors directly capture mechanical impacts and oscillations caused by bearing damage. Motor current is still valuable because it is non-invasive, but bearing faults appear more indirectly in current signals through torque ripple and electromagnetic coupling. Therefore, current-only diagnosis is usually more difficult and requires stronger feature engineering, more data, or deeper models.

The result also shows that physics-informed features are effective for the selected Paderborn subset. The feature-importance plot indicates that the model uses a combination of general spectral features and bearing-diagnostic harmonic features, which supports the intended design goal of combining signal-processing knowledge with trainable classification.

However, the current result should be interpreted as a progress checkpoint rather than a final claim for all operating conditions. The experiment uses a representative subset of four bearing archives and a capped number of windows per file. The next evaluation should include more bearing states, more operating conditions, and stricter train-test splits that separate operating conditions or bearing IDs.

12 Reproducibility Instructions

The following commands reproduce the current vibration-based result from the project root directory.

```
python -m pip install -e .[paderborn]

pinn-motor-fault download --codes K001 KA01 KI01 KB23 --extract

pinn-motor-fault train --data-dir data\paderborn\extracted --epochs 40 \
    --max-windows-per-file 16 --signal vibration \
    --output models\paderborn_vibration_pinn.npz

pinn-motor-fault results --data-dir data\paderborn\extracted \
    --model models\paderborn_vibration_pinn.npz --signal vibration \
    --max-windows-per-file 16 --output-dir reports\results_vibration
```

For a quick software validation without downloading the full dataset, the following smoke test can be used:

```
pinn-motor-fault smoke --epochs 10
```

13 Current Limitations

- The current high-accuracy result is based on a selected four-class subset, not the complete Paderborn dataset.
- The latest report uses vibration because it gives strong bearing-fault separability; motor-current-only diagnosis remains more challenging.

- The present split is window-based. A stricter future validation should split by operating condition or bearing state to test generalization more rigorously.
- The model is intentionally lightweight and implemented in NumPy. A CNN/PINN hybrid may improve robustness for larger-scale experiments.

14 Next Steps

The next phase of the project will focus on:

1. expanding from the four-class representative subset to more Paderborn bearing states;
2. comparing vibration-only, current-only, and combined current-vibration inputs;
3. evaluating generalization across load, speed, and torque conditions;
4. tuning the physics-regularization weight λ ;
5. implementing a deeper CNN/PINN hybrid for direct window-level feature learning;
6. preparing final project results with complete plots, ablation studies, and comparison against conventional machine-learning baselines.

15 Conclusion

At this stage, the project has progressed from an abstract concept to a complete working implementation. The pipeline can download and extract real Paderborn data, parse MATLAB sensor files, create signal windows, compute physics-informed features, train a neural classifier, evaluate performance, and generate report-ready visualizations. The current vibration-based model achieves **99.49%** accuracy on the evaluated four-class subset, with high precision and recall across all target classes. This provides a strong foundation for the next phase, where the focus will shift toward broader dataset coverage and stricter generalization testing.

References

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- [2] Paderborn University KAt Bearing DataCenter, “Bearing Data Center: Data Sets and Download.” Available: <https://mb.uni-paderborn.de/en/kat/research/bearing-datacenter/data-sets-and-download>
- [3] N. Herwig, P. Borghesani, W. Smith, and Z. Peng, “Signal processing- and physics-informed neural network for explainable bearing condition monitoring,” *Mechanical Systems and Signal Processing*, 2025.
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